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Assignment : 1 (CA - Software)

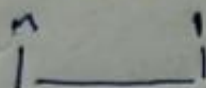
* MVC (Model - View - Controller) Design Pattern *

MVC is a software architectural pattern that separates an application into three interconnected components:

- * Model * Manages the data and business logic.
- * View * Displays the data (UI) to the user.
- * Controller * Handles user input and updates the Model and View accordingly.

Diagram

User → Controller → Model → View → User



Variants of MVC: *

* Model - View - Presenter (MVP): *

- Presenter handles all UI logic
- View is more passive; updates only via Presenter.
- Better for unit testing and windows desktop apps.

* Model - View - View Model (MVVM): *

- View Model binds the View and Model using data-binding
- Common in frameworks like Angular and WPF.
- Ideal for rich, interactive UIs.

When to Use: *

Use MVC for traditional web applications

Use MVP for test-driven desktop application

Use MVVM when UI binding and responsiveness are key.